Practice Set B

Class 12 Mathematics, Chapter 12: Linear Programming

Maximum marks: **27** **10** questions

Name _____ Date _____ Score _____ / 27

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

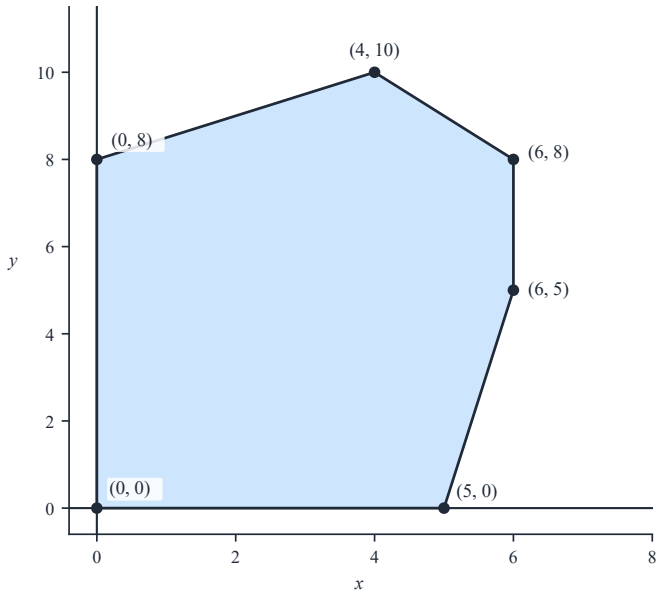
Q No.	Question	Marks

Level 1

4 marks

1 MCQ, NCERT Exemplar — 1 mark

The feasible region for an LPP is shown in the figure. Let $Z = 3x - 4y$ be the objective function. (Maximum value of Z + Minimum value of Z) is equal to:

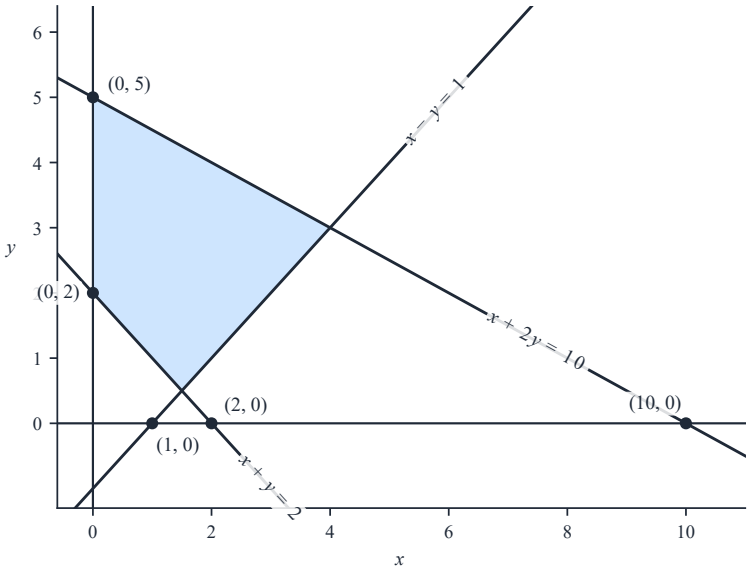


(A) 13 **(B)** 1 **(C)** -13 **(D)** -17

Q No.	Question	Marks

2 MCQ, CBSE SQP 2023-24 — 1 mark

The feasible region corresponding to the linear constraints of a Linear Programming Problem is given below. Which of the following is not a constraint to the given Linear Programming Problem?



(A) $x + y \geq 2$ **(B)** $x + 2y \leq 10$ **(C)** $x - y \geq 1$ **(D)** $x - y \leq 1$

3 MCQ, Board-pattern — 1 mark

For a linear programming problem with objective function $Z = ax + by$, which of the following statements is **false**?

(A) If the feasible region is unbounded, a maximum or a minimum value of Z may not exist.

(B) A corner point of a feasible region is a point in the region which is the intersection of two boundary lines.

(C) If the origin is a corner point of the feasible region, then the minimum value of Z is always 0.

(D) Points within and on the boundary of the feasible region represent feasible solutions.

4 Assertion-Reason, Board-pattern — 1 mark

Assertion (A): The linear programming problem: Maximise $Z = x + y$ subject to $x - y \leq -1, -x + y \leq 0, x \geq 0, y \geq 0$ has no optimal solution. **Reason (R):** The feasible region of this linear programming problem is unbounded.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Q No.	Question	Marks
Level 2		**9 marks**
5	Short answer, NCERT Exemplar Solve the following linear programming problem graphically: Minimise $Z = 13x - 15y$ subject to the constraints $x + y \leq 7, 2x - 3y + 6 \geq 0, x \geq 0, y \geq 0$.	3 marks
6	Short answer, CBSE SQP 2023-24 Solve the following Linear Programming Problem graphically: Maximise $Z = -x + 2y$ subject to the constraints $x \geq 3, x + y \geq 5, x + 2y \geq 6, y \geq 0$.	3 marks
7	Short answer, NCERT Exemplar Solve the following linear programming problem graphically: Maximise $Z = x + y$ subject to the constraints $x + 4y \leq 8, 2x + 3y \leq 12, 3x + y \leq 9, x \geq 0, y \geq 0$.	3 marks
Level 3		**14 marks**
8	Long answer, NCERT Exemplar Solve the following linear programming problem graphically: Maximise and minimise $Z = 3x - 4y$ subject to the constraints $x - 2y \leq 0, -3x + y \leq 4, x - y \leq 6, x \geq 0, y \geq 0$.	5 marks
9	Long answer, NCERT Exemplar A company manufactures two types of sweaters: type A and type B. It costs ₹360 to make a type A sweater and ₹120 to make a type B sweater. The company can make at most 300 sweaters and spend at most ₹72000 a day. The number of sweaters of type B cannot exceed the number of sweaters of type A by more than 100. The company makes a profit of ₹200 for each sweater of type A and ₹120 for every sweater of type B. Formulate this problem as an LPP and solve it graphically to find how many sweaters of each type the company should make in a day to get the maximum profit. Also find the maximum profit.	5 marks
10	Case study, NCERT Exemplar **Case Study:** A company makes three models A, B and C of calculators at two factories I and II. It has orders for at least 6400 calculators of model A, 4000 of model B and 4800 of model C. Every day, factory I makes 50 calculators of model A, 50 of model B and 30 of model C, while factory II makes 40 of model A, 20 of model B and 40 of model C. It costs ₹12000 a day to operate factory I and ₹15000 a day to operate factory II. Let factory I operate for x days and factory II for y days. **(i)** Write the objective function and the constraints of the LPP in simplest form. — 1 mark **(ii)** Find the corner points of the feasible region. — 1 mark **(iii)** Find the minimum operating cost, and justify why the minimum exists even though the feasible region is unbounded. — 2 marks	4 marks

Answer key

Chapter 12: Linear Programming, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) -17
2	(C) $x - y \geq 1$
3	(C) If the origin is a corner point of the feasible region, then the minimum value of Z is always 0.
4	(C) Assertion (A) is true, but Reason (R) is false.

Level 2

5	Minimum $Z = -30$ at $x = 0$, $y = 2$.
6	Z has no maximum value (the feasible region is unbounded and $-x + 2y > 1$ meets it).
7	Maximum $Z = \frac{43}{11}$ at $x = \frac{28}{11}$, $y = \frac{15}{11}$.

Level 3

8	Maximum $Z = 12$ at $(12, 6)$; Z has no minimum value.
9	150 sweaters of type A and 150 of type B a day; maximum profit ₹48000.
10	(i) Minimise $Z = 12000x + 15000y$ subject to $5x + 4y \geq 640$, $5x + 2y \geq 400$, $3x + 4y \geq 480$, $x, y \geq 0$ (ii) $(0, 200)$, $(32, 120)$, $(80, 60)$, $(160, 0)$ (iii) ₹18,60,000, with factory I for 80 days and factory II for 60 days